Sinan Li

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PROFILE

Research engineer working across formal mathematics and production systems. Trained at Oxford in *quantum error correction* and diagrammatic reasoning, then applied the same mathematics to **AI interpretability** at a stealth startup during the MSc. Separately co-founded *IQBank*, a course study platform serving **379 users**, and lead the team that builds and runs it.

EDUCATION

University of Oxford — 2025–2026
MSc in Advanced Computer Science (awaiting conferral) — *Oxford, United Kingdom*

- Dissertation: *The Effect of Fault-Equivalence Rewrites on Quantum Error Decodability*. Supervisor: Professor Aleks Kissinger.
- Areas: Quantum computing, formal verification, databases, uncertainty in deep learning, geometric deep learning.

University of Toronto — 2021–2025
Bachelor of Science, Computer Science Specialist; Mathematics Minor — *Toronto, ON, Canada*

- Graduated with **High Distinction**; GPA: **3.94 / 4.0**.
- Areas: Operating systems, machine learning, artificial intelligence, deep learning.

EXPERIENCE

Fabrial Labs Ltd. — Jun 2026–Aug 2026
Research Engineer, Early-Stage AI Research Startup (Stealth) — ***AI Interpretability Research***

- Applied *diagrammatic methods from category theory*, a formalism that came out of quantum computing, to modelling machine learning processes.
- Extended open-source **diagrammatic rewriting tools** with the group's own constructions, and wrote the **Python** verification code that tested theoretical claims against AI models and training methods.
- Validated the formalism symbolically against small **MLPs**, turning up counterexamples that sent the theory back.

IQBank — Apr 2025–Present
Co-founder, Development Lead & Product Manager — ***Product + Engineering Leadership***

- Built the study platform in **TypeScript**, React, and **PostgreSQL** for **379 users**. Remain its **largest code contributor** and run it in production on university infrastructure with **Docker** and **CI**.
- Designed the **access-control model**: orthogonal platform-wide (admin / staff / regular) and per-course (instructor / TA / student) role hierarchies.
- Lead the **10-person development team** directly and an education-research team through its lead, handle developer interviews and onboarding, and coordinate from the UK across Toronto working hours.
- Run the **user-research** loop that sets the roadmap: a colour-blind user reporting unreadable text led to a rebuilt colour system, and requests for personalisation became question bookmarks and personal notes.

University of Toronto — Jan 2025–Apr 2025
Teaching Assistant, CSC309: Programming on the Web — ***Academic Instruction***

- Owned one of the course's major assignments for a cohort of **300+ students**: designed it with the professor, then wrote the spec, rubric, reference solution, and **automated test suite** for a **RESTful** loyalty-program backend. It has been reused in later terms.
- Held weekly office hours across the full syllabus: **front-end and back-end web development**, **REST APIs**, and debugging.

PROJECTS

KidneyOS: Rust-Based Toy Operating System — 2024
University of Toronto; supervised by Professor Kuei Sun — ***Rust + Systems***

- Principal author of the **storage subsystem** below the VFS layer in a from-scratch **Rust** kernel built by eight people, and sole author of its **PS/2 keyboard driver**.
- Wrote an **ATA driver** whose interrupt handlers wake semaphore-blocked threads, a **block-device layer** that hid hardware specifics from the filesystem above, and **MBR partition** parsing and registration.
- Built the disk-image tooling the kernel boots against, generating partitioned images formatted as **FAT16, ext2, or ext4**.

Evogami: 3D Puzzle Platformer — 2024

University of Toronto and Sheridan College — ***Programming Leadership***

- **Programming Lead** on a 12-person team of computer science, art, and music students across two institutions, shipping an original game in **Unity** for Windows and macOS.
- Wrote **94% of the game's C# gameplay code**: the origami transformation state machine behind the core mechanic, character movement with ground and wall detection, the UI, and an input layer covering keyboard, mouse, and **Xbox and PlayStation controllers**.
- Directed the programming team and worked out technical decisions with the design and audio groups.

AWARDS & HONOURS

Dean's List Scholar | University of Toronto — **2021–2025**

Later Life Learning Scholarship | University of Toronto — **2021, 2024**

Innis College Exceptional Achievement Award | University of Toronto — **2023**

TECHNICAL SKILLS

Languages: Python, TypeScript, Rust, C#, C/C++, Java, SQL, Bash
ML & AI: AI interpretability, deep learning, geometric deep learning, uncertainty quantification, PyTorch, NumPy
Research Areas: Quantum computing, quantum error correction, ZX-calculus, category theory, formal verification
Web & Backend: React, Node.js, Express.js, PostgreSQL, Prisma ORM, REST APIs, ASP.NET
Systems: Operating systems, kernel development, device drivers, concurrency, x86
Tools & Platforms: Linux, Docker, CI, AWS S3, Unity
Engineering: Technical leadership, experiment design, user research, product iteration